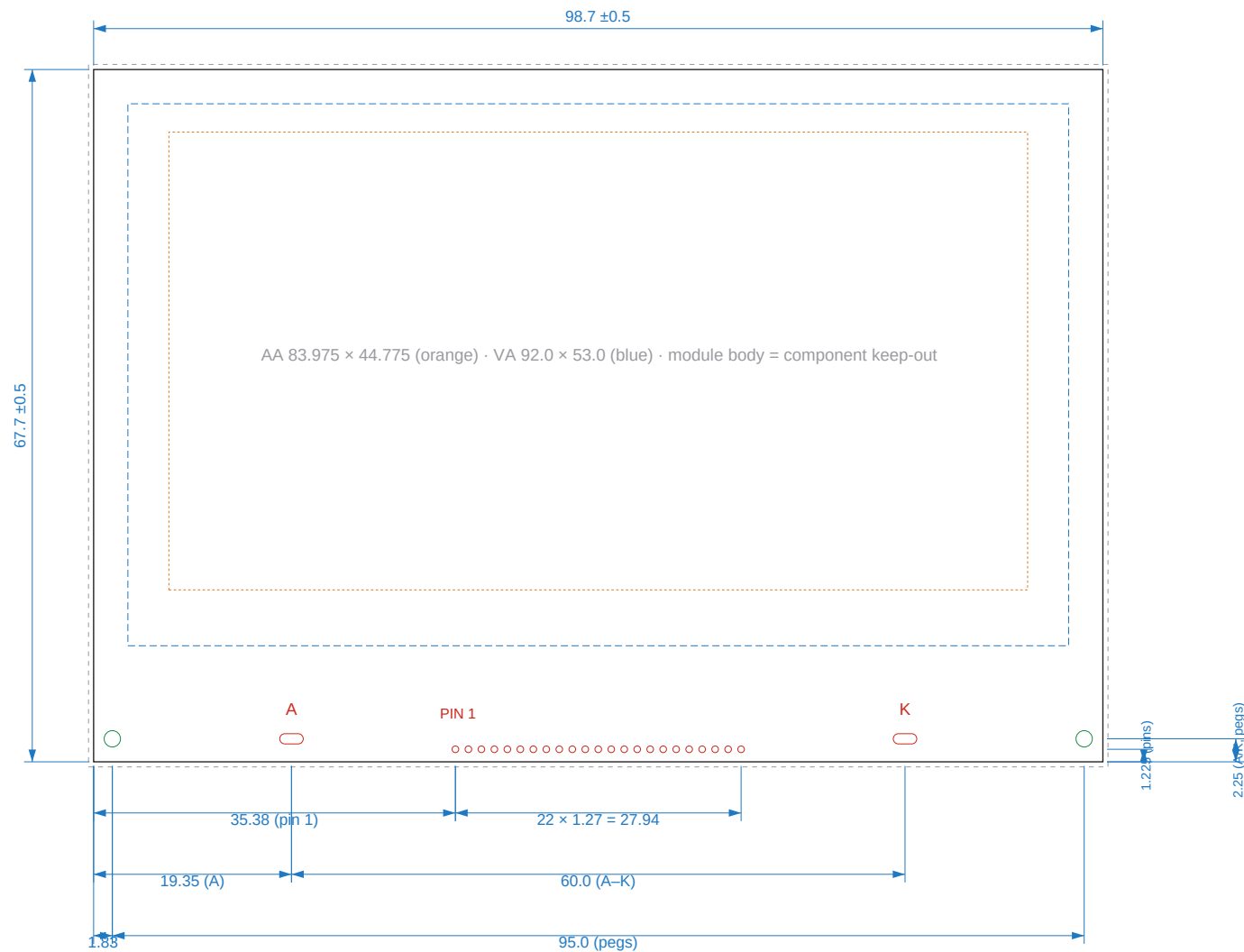




Holes

23 × signal	PTH Ø0.65, pad 1.0×1.6 oval (pin 1 rect) lead 0.40 × 0.25, P1.27, Y = 1.225
2 × LED A / K	PTH slot 2.3×1.0, pad 3.0×1.6 tab 1.8 × 0.5, Y = 2.25
2 × pegs Ø1.5	NPTH Ø1.6, Y = 2.25
Courtyard	outline + 0.5
Body	sits flat on the PCB: keep-out under it

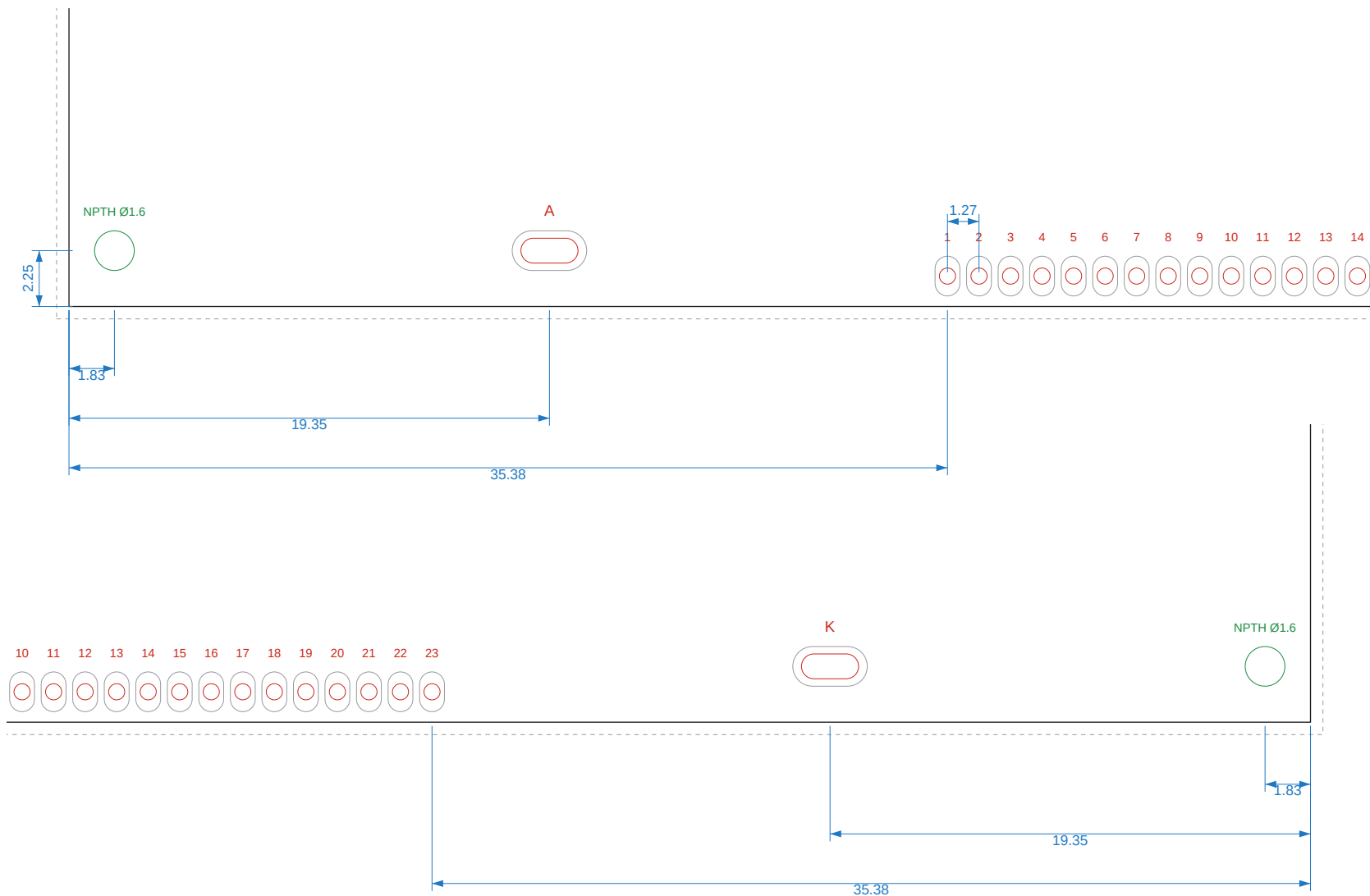
Exact coordinates: holes.csv · drill/pad sizes are recommendations; outline, pin and peg positions are from the Winstar drawing.

WO240128A — PCB footprint

Hole pattern for the module's bent pins, LED tabs and locating pegs

PerchWerks datalogger · WO240128A · mm · scale 1.5:1 · sheet 1/5

Front / PCB top view. Origin: outline bottom-left. Generated by src/make_handoff.py



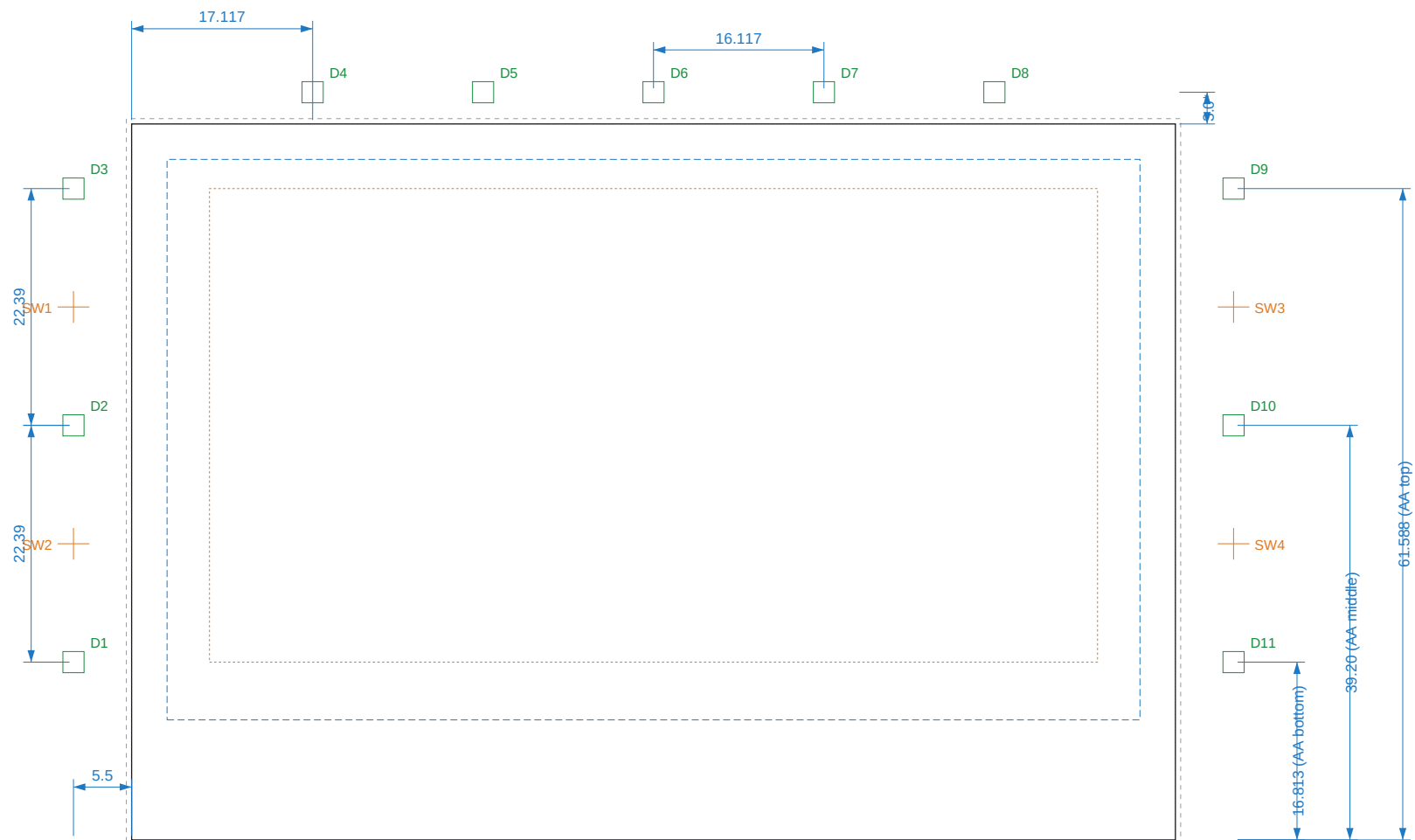
Signal-pin row at Y = 1.225; A/K tabs and pegs at Y = 2.25.

Grey = recommended pads. Signal: Ø0.65 drill, 1.0×1.6 oval (0.27 copper gap at P1.27). LED tabs: 2.3×1.0 slot, 3.0×1.6 pad. Pegs: Ø1.6 NPTH.

Detail — pin row, LED tabs, pegs

Bottom edge of the module, enlarged

PerchWerks datalogger · WO240128A · mm · scale 4:1 (two halves) · sheet 2/5
Front / PCB top view. Origin: outline bottom-left. Generated by src/make_handoff.py



Top row: 5 LEDs on a 16.1167 pitch, centred (the pitch of 7 spanning the outline edge to edge; the outer two are not fitted).

Side columns (LEDs and buttons) 5.5 mm out from the outline, level with the AA top / middle / bottom; 22.39 c-c, 20.39 clear for the buttons.

SW1–SW4: suggested button centres, midway between side LEDs. Chain: up the left column, along the top, down the right.

* placeholder until the enclosure is drawn: top-row gap above the outline (3.0). The side offset (5.5) is confirmed.

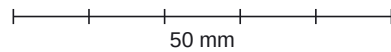
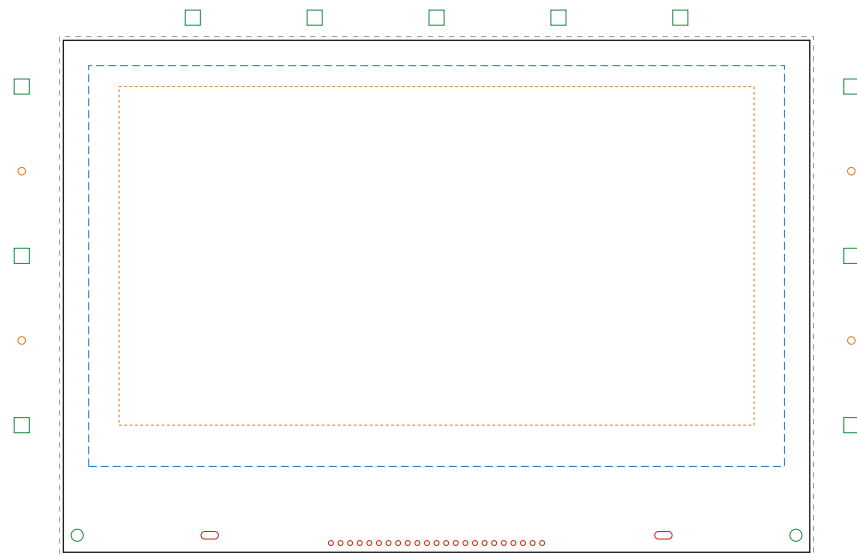
The LEDs sit on the PCB, 5.7 mm below the glass front face: plan light pipes or a stepped bezel. Full list: leds.csv.

LED + button placement — 5 top + 3 + 3 (confirmed)

11 × WS2812B-2020 on the PCB top side, D1–D11 in data-chain order

PerchWerks datalogger · WO240128A · mm · scale 1.6:1 · sheet 3/5

Front / PCB top view. Origin: outline bottom-left. Generated by src/make_handoff.py



1:1 check print

Print at 100 % (no fit-to-page) and check the 50 mm bar, then lay the module or PCB on it

PerchWerks datalogger · WO240128A · mm · scale 1:1 · sheet 4/5

Front / PCB top view. Origin: outline bottom-left. Generated by src/make_handoff.py

Module: Winstar WO240128A, 240 × 128 COG (UC1608), outline 98.7 × 67.7 ±0.5, 9.5 overall (to the A/K tips).

Mounting: the module lies flat on the PCB top side, frame rear face on the board; pins, A/K tabs and pegs go through.

Glass front face is 5.7 above the PCB (2.9 glass + 2.8 backlight). The whole 98.7 × 67.7 body is a component keep-out on top.

Lead lengths below the frame rear face: signal pins 5.2, A/K tabs 3.8, pegs 2.0 — trim or allow for them under a 1.6 PCB.

Pin 1 is the leftmost signal pin seen from the front (X 35.38). Check the pin-out table on the datasheet revision you buy.

From the drawing: outline, pin positions / pitch, A/K and peg positions. Estimated: A/K tab thickness, chip size.

Recommended (ours): drill and pad sizes — signal Ø0.65 / 1.0×1.6 oval, A/K slot 2.3×1.0 / 3.0×1.6, pegs NPTH Ø1.6.

If your fab's minimum annular ring or slot rules differ, keep the hole centres and adjust sizes.

LEDs: 11 × WS2812B-2020 (2.0 × 2.0 × ~0.84), chain D1→D11 = up the left column, along the top, down the right.

Layout confirmed: 5 on top, 3 per side; side LEDs and buttons 5.5 out from the outline. Roles in firmware: top row = pace, side columns = status (overheat, rev limit, grip), all = purple sector / best lap.

Budget ~15 mA per LED at full white (5 mA/colour 2020 types; some are 12 mA/colour — check the part), ~0.5–1 mA idle each.

Consider a load switch on the LED 5 V rail so firmware can cut the idle current, and a 3.3 V→5 V level shifter on DIN.

Files: holes.csv (from pin 1 and from the corner), leds.csv, buttons.csv, WO240128A_pcb.dxf (layered),

WO240128A_COG.kicad_mod (origin pin 1), plus ../scad (OpenSCAD source: footprint / bezel modes) and ../exports (STL, DXF).

Regenerate after any change: python3 src/make_handoff.py (reads the .scad).

Notes for layout

What the drawing fixes, what is ours to choose

PerchWerks datalogger · WO240128A · mm · — · sheet 5/5

Front / PCB top view. Origin: outline bottom-left. Generated by src/make_handoff.py